PRACTICAL-07

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
int global_var = 100;
static int static_var = 200;
void show_addresses()
{
   int local_var = 300;
   printf("\n===== MEMORY ADDRESSES =====\n");
   printf("Code/Text address   : %p\n", (void *)show_addresses);
   printf("Global variable     : %p\n", (void *)&global_var);
   printf("Static variable     : %p\n", (void *)&static_var);
   int *heap_var = malloc(sizeof(int));
   if (heap_var == NULL)
   {
      printf("Heap allocation failed\n");
      return;
   }
   *heap_var = 400;
   printf("Heap variable       : %p\n", (void *)heap_var);
   printf("Stack variable      : %p\n", (void *)&local_var);
   printf("\nProcess ID (PID)    : %d\n", getpid());
   free(heap_var);
}
int main()
{
   show_addresses();

   printf("\nPress Enter to exit...\n");
   getchar();
   return 0;
}
```

Screen Shots:

```
sohan_jadala@DESKTOP-OI7PGP2:~/2520080060_OSSP/2520080060_Practical$ gedit Practical_7.c

** (gedit:565): WARNING **: 06:49:26.641: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.
0' not found

** (gedit:565): WARNING **: 06:49:26.641: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', versi
on '1.0' not found
sohan_jadala@DESKTOP-OI7PGP2:~/2520080060_OSSP/2520080060_Practical$ gcc Practical_7.c -o Practical_7
sohan_jadala@DESKTOP-OI7PGP2:~/2520080060_OSSP/2520080060_Practical$ ./Practical_7

===== MEMORY ADDRESSES =====
Code/Text address   : 0x5a06623e8209
Global variable     : 0x5a06623eb010
Static variable     : 0x5a06623eb014
Heap variable       : 0x5a06675c3420
Stack variable      : 0x7ffe805c24fc

Process ID (PID)    : 1289

Press Enter to exit...
cpu
sohan_jadala@DESKTOP-OI7PGP2:~/2520080060_OSSP/2520080060_Practical$
```